Extracting the Born cross section for $ep \to e'p'J/\psi$ at CLAS12: procedure, master formula, and the traps in our own samples

Campaign `prod_jpsi_f18in_20261009`

`exclurad_py` note, branch `jpsi-born-xsec-note`

6 November 2026

Abstract

What to do with a data yield binned in (W, Q^2, t), a radiative Monte Carlo and a Born Monte Carlo in order to obtain a Born cross section. The master formula is given in two algebraically equivalent forms. The one to *compute* is the ratio of the data yield to the reconstructed radiative Monte Carlo, times the model: in that form a wrong model *normalisation* cancels exactly, which mattered for the superseded `jpsi.diffrad_vm`, which sat a factor 4–6 below the data. Only the shape inside the bin survives, and it is iterated away. Six inputs beyond the three samples are required and are listed explicitly; the branching ratio $\mathcal{B}(J/\psi \to e^+e^-)$ is one of them and is *not* applied anywhere in the generator, a factor 16.7 if missed. Moreover the PDG branching ratio is the *radiation-inclusive* leptonic width, so radiation of the decay leptons belongs to the final state that $\mathcal{B}$ already counts and must not be allowed into C^{rad}: the Born arm has to be the `borndecay` one, with Born production and decay radiation on (Sec. 5.2). That premise is checked against the PDG listing, which quotes the radiative submode separately for $E_\gamma > 100$ MeV: the generator's decay model gives $14.38 \pm 0.06\,\%$ against the PDG's $14.74 \pm 2.35\,\%$, agreeing to 0.15σ (Sec. 5.3). Several properties of the existing campaigns were verified while writing this note and are reported in Sec. 5, including the card fields of all three arms and the fact that the generator's `sigma_nb` cannot be used to form the radiative correction.

1 The inputs

Three samples are not enough. The complete set is Table 1. Items 4–10 are not refinements: each is a factor, or a bias of the order of the effect being measured.

#	Input	Source
1	data yield in (W, Q^2, t) bins	analysis ntuple
2	radiative MC through GEMC	`osg_mariana_lund`, arm `rad`
3	**borndecay** MC through GEMC	`osg_mariana_lund`, arm `borndecay`; Sec. 5.2
4	integrated luminosity $\mathcal{L}$	for exactly the run/trigger selection that made the yield
5	signal extraction prescription	$M(ee)$ window and background shape
6	$\mathcal{B}(J/\psi \to e^+e^-) = (5.971 \pm 0.032)\,\%$	PDG. **Not in the generator**, Sec. 5.1
7	data/MC efficiency scale factors	FT electron, lepton and proton tracking, trigger
8	bin volumes, centroids, averaging rule	chosen once and written down
9	one cut list for data and *both* MC arms	exclusivity cut included
10	a model to iterate with	Sec. 3, step 7

Table 1: Inputs to the extraction.

2 The master formula

Each arm carries a per-event weight built from what the generator already reports in `<job>.config.json`,

$$w^{\mathrm{arm}} = \frac{\sigma^{\mathrm{arm}}_{\mathrm{gen}}}{N^{\mathrm{arm}}_{\mathrm{gen}}} \quad [\mathrm{nb/event}], \qquad \sigma_{\mathrm{gen}} \equiv \texttt{report.sigma_nb}. \tag{1}$$

Let n_i be the number of *reconstructed* Monte Carlo events of a given arm falling in bin i after the identical analysis chain, and m_i the number of *generated* (truth) events in bin i with no cuts at all.

Form A — ratio to the radiative MC. This is the one to compute. R_i is a ratio of two *radiative-level* numbers: numerator and denominator are both reconstructed, after cuts, with radiation present, because the data are radiative and the comparison must be like with like. There is no Born quantity in R_i at all — the only Born object in Form A is the model factor multiplying it.

$$\boxed{\sigma_i^{\mathrm{Born}} = R_i\, \bar{\sigma}_i^{\mathrm{Born,model}}, \qquad R_i = \frac{N_i^{\mathrm{data}}}{\mathcal{L}\,\mathcal{B}\, w^{\mathrm{rad}}\, n_i^{\mathrm{rad}}}} \tag{2}$$

Reading Form A factor by factor. The denominator is not an abstraction: it is the number of events we expect to see.

factor	what it is	units
w^{rad}	cross section per generated event, $\sigma_{\mathrm{gen}}/N_{\mathrm{gen}}$	nb/event
n_i^{rad}	RECONSTRUCTED rad-arm events in bin i after all cuts	events
$w^{\mathrm{rad}} n_i^{\mathrm{rad}}$	cross section of the phase space reconstructing into bin i	nb
$\mathcal{B}$	$\mathcal{B}(J/\psi \to e^+e^-) = 5.971\,\%$	—
$\mathcal{L}$	integrated luminosity of the data	nb^{-1}

So $\mathcal{L}\,\mathcal{B}\, w^{\mathrm{rad}} n_i^{\mathrm{rad}}$ is the *expected number of events* in bin i if the model were right. Dimensions: $\mathrm{nb}^{-1}\times\mathrm{nb}$ = events, so R_i is events/events — dimensionless — and σ_i^{Born} inherits its units from $\bar{\sigma}_i^{\mathrm{Born,model}}$, namely nb. $\mathcal{B}$ enters here exactly once, because the generator decays *every* event with probability one (Sec. 5.1): σ_{nb} is the full production cross section while the data yield is an e^+e^- yield.

R_i alone says only "the data are N times the model" — a deviation, not a cross section. Multiplying by the model's Born cross section in the same bin turns the deviation into an absolute number: *the model is a carrier of units and shape, not a prediction of the magnitude.* And the denominator is the *radiative* expectation because the data are radiative; the `borndecay` arm is needed only for C^{rad} as a diagnostic and for Form B.

Form B — explicit factors. This is the one to tabulate systematics on.

$$\bar{\sigma}_i^{\mathrm{Born}} = \frac{N_i^{\mathrm{data}}}{\mathcal{L}\,\mathcal{B}\,\Delta\Omega_i\,\varepsilon_i^{\mathrm{Born}}\,C_i^{\mathrm{rad}}}, \qquad \varepsilon_i^{\mathrm{Born}} = \frac{n_i^{\mathrm{bdec}}}{m_i^{\mathrm{bdec}}}, \qquad C_i^{\mathrm{rad}} = \frac{w^{\mathrm{rad}}\, n_i^{\mathrm{rad}}}{w^{\mathrm{bdec}}\, n_i^{\mathrm{bdec}}}. \tag{3}$$

Here `bdec` is the **borndecay** arm: Born production with decay radiation *on*. It is not the `born` arm. Section 5.2 gives the reason, and it is the one choice in this note that cannot be got wrong without spoiling the answer.

What Form A assumes, and what it does not

Write the whole chain — radiation, resolution, migration, acceptance, cuts — as one operator K that takes a *Born* cross section in and returns a *reconstructed radiative* yield. Then

$$N_i^{\text{data}} = \mathcal{L}\,\mathcal{B}\,(K\sigma^{\text{true}}_{\text{Born}})_i, \qquad \mathcal{L}\,\mathcal{B}\,w^{\text{rad}} n_i^{\text{rad}} = \mathcal{L}\,\mathcal{B}\,(K\sigma^{\text{model}}_{\text{Born}})_i, \tag{4}$$

so R_i is the ratio of two *outputs* of K, and it factorises:

$$R_i = \frac{(K\sigma^{\text{true}})_i}{(K\sigma^{\text{model}})_i} = \underbrace{\frac{\sigma^{\text{true}}_{\text{Born}}}{\sigma^{\text{model}}_{\text{Born}}}}_{\text{wanted}} \times \underbrace{\frac{C^{\text{rad}}_{\text{true}}}{C^{\text{rad}}_{\text{model}}}}_{\text{assumed=1}} . \tag{5}$$

R_i is therefore the Born ratio *only if* $C^{\text{rad}}_{\text{true}} = C^{\text{rad}}_{\text{model}}$. That is the assumption of the method, and it is not free: it is the physics content of the RC engine, tested by the two-method comparison of `docs/rc_method_jpsi/` and by the generator-level closure to 0.1 %, not by this formula.

Equivalently, Form A is exact if and only if $\sigma^{\text{true}}/\sigma^{\text{model}}$ is *constant* over the region feeding bin i — the bin plus the migration tails plus the radiative smearing. K is then linear there, $(Kc\sigma)_i = c(K\sigma)_i$, and $R_i = c$ exactly. Step 7 is what makes that hold: after reweighting, the ratio is locally constant by construction. So the overall normalisation of the model cancels exactly, while the model's *shape* inside the bin and the correctness of C^{rad} do not.

Where the assumption comes from: the structure of the RC formula

The driver implements (`rc.py:5`; Akushevich Eq. 55)

$$\sigma_{\text{obs}} = \sigma_{\text{Born}}\, e^{\delta_{\text{inf}}}\,(1 + \delta_{VR} + \delta_{\text{vac}}) \;+\; \sigma_F, \qquad \eta = \frac{\sigma_{\text{obs}}}{\sigma_{\text{Born}}}, \tag{6}$$

a *multiplicative, local* factor plus an *additive, non-local* integral — not σ_{Born} times a number. Measured over the J/ψ box with `jpsi.vmd2026_pdet` and no cut:

Q^2	W^2	t	ϕ	σ_{Born}	virt. part	σ_F	η	$\sigma_F/\sigma_{\text{obs}}$
0.03	17.50	−1.50	0	6.73314e-05	6.36440e-05	−1.2473e-06	0.9267	−0.0200
0.03	19.50	−1.50	0	8.36546e-05	8.57950e-05	−3.8327e-06	0.9798	−0.0468
0.03	19.50	−1.50	π	8.36546e-05	8.55674e-05	−3.7141e-06	0.9785	−0.0454
0.03	19.50	−3.00	0	1.85470e-05	1.90773e-05	−8.8572e-07	0.9808	−0.0487
1.00	19.00	−1.50	0	1.85792e-06	1.88513e-06	−8.9031e-08	0.9667	−0.0496

The multiplicative factor sits slightly *above* unity (vacuum polarisation and virtual), while σ_F is *negative*, 2–5 % of σ_{obs}; η is the difference of the two.

Eq. (6) *proves* the normalisation invariance rather than threatening it. Both terms are linear and homogeneous of degree 1 in the structure functions — σ_{Born} is linear in them and σ_F is an *integral* of them — so SF $\to f$ SF scales both by f and η cannot move:

f	σ_{Born}	virt. part	σ_F	η
1.00	8.365462e-05	8.579501e-05	−3.83270e-06	0.979770256397
2.00	1.673092e-04	1.715900e-04	−7.66539e-06	0.979770256397
0.25	2.091365e-05	2.144875e-05	−9.58174e-07	0.979770256397
5.00	4.182731e-04	4.289750e-04	−1.91635e-05	0.979770256397
1000	8.365462e-02	8.579501e-02	−3.83270e-03	0.979770256397

Twelve digits over a factor 4000. Eq. (19) is therefore a property of the RC formula, not an assumption. Eq. (6) also *identifies the region*. The line that matters is `radiative.py:246`,

```
sf = _sf_to_gev(sfmodel(W2t, np.maximum(Q2t, 1e-6), tt))
```

σ_F reads the structure functions at *shifted* W^2, Q^2 and t inside the photon integral. So η at an observed point is not a function of the model at that point: it is a functional of the model's *shape* over the region the photon integral reaches. Hence, exactly: the normalisation cancels by degree-1 homogeneity, while the shape over the σ_F region does not — and that is $C^{\text{rad}}_{\text{true}} = C^{\text{rad}}_{\text{model}}$.

How σ_F depends on σ_{Born}

σ_F is a weighted integral of σ_{Born} *itself*, read at *shifted* invariants, minus a subtraction that reads σ_{Born} at the observed point:

$$\sigma_F = \int_0^{v_{\max}} dv \int d\Omega_k \; w \Big[\underbrace{R(v,\Omega_k)\,\sigma_{\text{Born}}(\tilde{W}^2,\tilde{Q}^2,\tilde{t})}_{\text{real emission, SHIFTED}} - \underbrace{F_{\text{IR}}(v,\Omega_k)\,\sigma_{\text{Born}}(W^2,Q^2,t)}_{\text{Bardin–Shumeiko, OBSERVED}} \Big], \tag{7}$$

with the shifts exactly as `radiative.py:239--243` writes them,

$$\tilde{W}^2 = W^2 - v, \qquad \tilde{Q}^2 = Q^2 + 2(k\cdot q), \qquad \tilde{t} = t - 2\big(k\cdot(q - p_V)\big). \tag{8}$$

So σ_F depends on σ_{Born} *twice*: through σ_{Born} at *other* kinematic points — the non-local part — and through σ_{Born} at the observed point with a minus sign, the subtraction that cancels the infrared divergence of the first term. Both are linear in σ_{Born}, so

$$\sigma_{\text{obs}}(x) = \underbrace{\sigma_{\text{Born}}(x)\,e^{\delta_{\text{inf}}}(1+\delta_{VR}+\delta_{\text{vac}})}_{\text{local}} + \underbrace{\int K(x \leftarrow x')\,\sigma_{\text{Born}}(x')\,dx'}_{\text{NON-LOCAL}} - \underbrace{C(x)\,\sigma_{\text{Born}}(x)}_{\text{local (BS)}}. \tag{9}$$

Three terms, every one linear and homogeneous of degree 1 in σ_{Born}. That is the whole structural answer: homogeneity makes η scale-free, the middle term makes η shape-dependent.

Degree-1 homogeneous means

$$F[c\,\sigma_{\text{Born}}] = c\,F[\sigma_{\text{Born}}] \qquad \text{for every constant } c, \tag{10}$$

i.e. multiply the whole Born cross section by c and the answer comes out multiplied by exactly c: no c^2, no c^0, no additive piece left over. It holds here because none of the kernels —$e^{\delta_{\text{inf}}}$, $(1+\delta_{VR}+\delta_{\text{vac}})$, K, C— contains σ_{Born}; they are pure QED and kinematics. So σ_{obs} scales by c and the c cancels in $\eta = \sigma_{\text{obs}}/\sigma_{\text{Born}}$.

It is *weaker* than proportionality, and the standard counter-example is exactly σ_F: take $F[\sigma](x) = \sigma(x+a)$. Scaling σ by c scales F by c, so F is degree-1 homogeneous, yet

$$\frac{F[\sigma](x)}{\sigma(x)} = \frac{\sigma(x+a)}{\sigma(x)} \tag{11}$$

depends on the *shape* of σ, not on its size — which is σ_F in one line, since it reads σ_{Born} at $\tilde{W}^2 = W^2 - v$ rather than at W^2.

Geometrically: discretise the kinematics, so that σ_{Born} is a vector and the radiative correction a matrix M independent of it,

$$\vec{\sigma}_{\text{obs}} = M\,\vec{\sigma}_{\text{Born}}, \qquad \eta_i = \frac{(M\vec{\sigma})_i}{\sigma_i}, \tag{12}$$

and η depends on the *direction* of σ_{Born} in function space, never on its *length*. Normalisation is the length; shape is the direction.

What would break it: an additive term carrying its own absolute normalisation, $F = M\sigma + b$ with b fixed in nb, is not degree-1 homogeneous, and then η would depend on the model's normalisation. Eq. (6) has no such term — every piece is proportional to the structure functions. A σ^2 term would be degree 2 and would make η scale linearly with the normalisation.

Which $\sigma_{\rm Born}$ values enter. At $Q^2 = 0.03$, $W^2 = 19.5$, $t = -1.5$ the limit is $v_{\max} = 2.065$, so $\tilde{W}^2$ spans $[17.435, 19.500]$; threshold (16.283) is not reached. Cumulating σ_F from $v = 0$ up to v_m:

v_m	$\tilde{W}^2$ at v_m	cum. σ_F	slice	slice/total
0.010	19.490	−4.18559e-08	−4.18559e-08	0.011
0.050	19.450	−1.96042e-07	−1.54186e-07	0.040
0.150	19.350	−5.28354e-07	−3.32312e-07	0.087
0.400	19.100	−1.18827e-06	−6.59918e-07	0.172
1.000	18.500	−2.33841e-06	−1.15014e-06	0.300
2.065	17.435	−3.83270e-06	−1.49429e-06	0.390

69 % of σ_F comes from $v > 0.4$, i.e. $\tilde{W}^2$ below 19.1 — a GeV^2 or more from the observed point — and 39 % from $v > 1$. That is the non-locality, measured. The $\sigma_{\rm Born}$ curve being weighted:

v	$\tilde{W}^2$	$\sigma_{\rm Born}$	/ observed
0.000	19.500	8.3655e-05	1.000
0.400	19.100	8.0646e-05	0.964
1.000	18.500	7.6085e-05	0.910
2.065	17.435	6.6633e-05	0.797

Why σ_F is negative here. $\sigma_{\rm Born}$ falls to 80 % of its observed value at the far end, because for J/ψ near threshold $\sigma(W)$ *rises*, so shifting W^2 down lowers the cross section. The real-emission term therefore reads smaller values than the subtraction term, which reads the full observed $\sigma_{\rm Born}$: the subtraction wins and $\sigma_F < 0$.

That is exactly what flipped in the shape test above. With an artificially *falling* $\sigma(W^2)$ ($a = -0.6$) the shifted $\sigma_{\rm Born}$ exceeded the observed one, the real term won, σ_F came out $+4.3 \times 10^{-6}$ and η rose above unity — radiation feeding events *into* the point from higher W^2. The *sign* of σ_F is therefore governed by the slope of $\sigma_{\rm Born}$ in W^2, which is also why the near-threshold W rise is the dominant shape freedom in the table above.

One consequence to record: $C^{\rm rad}$ cannot be derived from the Born cross section at the bin centre; it is an integral, so interpolating it in (W, Q^2, t) is bookkeeping, not derivation.

Retracted: an earlier revision added a second consequence, that σ_F samples the model below its Q^2 fit range 0.015–0.1 GeV^2. That range belongs to the *superseded* `jpsi.diffrad_vm`. The production model `jpsi.vmd2026` declares validity $W^2 \in (16.4, 400)$, $Q^2 \in (0, 30)$, $|t| \in (0, 12)$, so there is no Q^2 floor problem for J/ψ at all.

Inside the integral: what `exact - sub` is

The bracket above is literally three lines of `radiative.py:252--255`:

```
fir   = F_IR(Q2, m, kap1, kap2)
exact = LRH / Q2t**2              # full bremsstrahlung, shifted kinematics
sub   = 4.0 * fir * LH0 / Q2**2   # its own soft limit, Born kinematics
integrand = (v_f / z**2) * (exact - sub)
```

`exact` is the true one-photon integrand: the radiative leptonic tensor contracted with the hadronic tensor, `LRH`, over the *shifted* propagator $\tilde{Q}^4$, every ingredient evaluated at the kinematics the event has *after* emitting k.

`sub` is the same quantity in the soft limit $\omega \to 0$ and nothing more. There the leptonic tensor factorises, $L_R \to 4 F_{\rm IR} L_0$, which is the definition of $F_{\rm IR}$ (`spinors.py:200`),

$$F_{\rm IR} = \frac{Q^2 + 2m^2}{\kappa_1 \kappa_2} - \frac{m^2}{\kappa_1^2} - \frac{m^2}{\kappa_2^2}, \qquad \kappa_i = 2k \cdot k_i, \tag{13}$$

and the hadronic side is frozen at the Born point: `LH0` and Q^2, not $\tilde{Q}^2$. So `sub` is "the Born cross section times the eikonal photon factor".

Why subtract. Both terms behave as $1/\omega^2$ as $\omega \to 0$, and since $\omega = v/2z$ the measure $v\,dv/z^2$ turns that into dv/v: each one alone is logarithmically infrared divergent. Their *difference* is finite, because `exact` $\to$ `sub` exactly in the soft limit — a soft photon cannot resolve the shifts $\tilde{W}^2, \tilde{Q}^2, \tilde{t}$, and $F_{\rm IR}$ is built to be precisely that limit. The integrand then vanishes as $v \to 0$ instead of blowing up, and Gauss–Legendre in $\ln v$ converges.

The code's v grid starts at $\ln v_{\max} - 18$, an arbitrary number, and it is *allowed* to be arbitrary only because of the subtraction. Measured at $Q^2 = 1$ GeV^2, $W^2 = 19$ GeV^2, $t = -1.5$ GeV^2 with `jpsi.vmd2026`, moving that lower edge:

ln-span below $v_{\max}$	`subtract=True`	`subtract=False`
6	1.72232×10^{-8}	7.49972×10^{-7}
10	1.72100×10^{-8}	1.23846×10^{-6}
14	1.72087×10^{-8}	1.72696×10^{-6}
18	1.72074×10^{-8}	2.21545×10^{-6}
26	1.72043×10^{-8}	3.19245×10^{-6}

The right column rises by $+4.885 \times 10^{-7}$ per four units of $\ln v$ — dead linear, which *is* the logarithmic divergence, in the open. The left column is stable to 1.1×10^{-3} while the cutoff moves over twenty orders of magnitude.

Where the subtracted piece goes. It is not thrown away. $\int dv\, d\Omega_k\, (v/z^2)\, 4F_{\rm IR} L H_0/Q^4$ is doable in closed form, and the result is exactly $\delta_{\rm inf} = (\alpha/\pi)(l_m - 1)\ln\!\left(v_m^2/S_0' X_0'\right)$ (plus the soft part that pairs with the vertex loop inside δ_{VR}). Its infrared divergence cancels against the virtual loops, which is why Eq. (6) has the shape it has: soft + virtual analytically, as an IR-free sum, and the hard remainder σ_F numerically. So (`exact` − `sub`) and $\delta_{\rm inf}$ are two halves of one identity, and *no soft-photon cutoff parameter survives into the answer* — $v_{\max}$ there is a physical missing-mass cut, not a regulator. That is the content of the covariant Bardin–Shumeiko separation, as against the older peaking and cutoff schemes.

The identity is enforced, not assumed: the difference of `sigma_F` taken with `subtract=False` and `subtract=True` over the same $[v_{\rm lo}, v_{\rm hi}]$ must equal $\sigma_{\rm Born}\left[\delta_{\rm inf}(v_{\rm hi}) - \delta_{\rm inf}(v_{\rm lo})\right]$ (`validation/selftest.py:122`). Run on 6 November 2026:

```
soft factorization L_R -> 4 F_IR L_0              4.254e-04    OK
IR bookkeeping (subtraction == d delta_inf)       9.889e-05    OK
slicing form vs Bardin-Shumeiko                   3.372e-03    OK
```

`subtract=False` — **the slicing form, which is what the generator uses.** The subtracted integrand is *not* positive definite: the subtraction cancels the collinear divergence pointwise, so the bracket changes sign and cannot be a probability density. For generation the same total is reorganised at an auxiliary slicing point $v_{\min}$ (`generator/tail.py`):

$$\sigma_{\rm hard}(v_{\min}) = \int_{v_{\min}}^{v_{\max}} dv\, d\Omega_k\, \frac{v}{z^2}\, \frac{L_R : H}{\tilde{Q}^4} \;\geq 0 \qquad \text{[the sampling density]}, \tag{14}$$

$$\sigma_{\rm soft}(v_{\min}) = \sigma_{\rm obs}^{\rm BS} - \sigma_{\rm hard}(v_{\min}) \qquad \text{[virtual, soft real, and the unresolved } v < v_{\min} \text{ slice]}. \tag{15}$$

$\sigma_{\rm soft}$ is *defined* as the remainder of the exact Bardin–Shumeiko $\sigma_{\rm obs}$, so soft + hard = $\sigma_{\rm obs}$ identically for any $v_{\min}$: both organisations give the same number, split differently. The trap, recorded in that file, is

the naive $\sigma_{\rm soft} = \sigma_{\rm Born} e^{\delta_{\rm inf}(v_{\rm min})}(1+\delta_{VR}+\delta_{\rm vac})$, which re-exponentiates at an arbitrary point and double-counts: at a typical CLAS12 η point $\delta_{\rm inf}(v_{\rm max}) = -0.26$ against $\delta_{\rm inf}(10^{-6}v_{\rm max}) = -1.13$, giving $\sigma_{\rm obs}$ a $\sim$20 % dependence on a number that is not physics. The exponentiation must stay at $v_{\rm max}$ and only the IR bookkeeping may cross $v_{\rm min}$. Nor is $v_{\rm min}$ free: $\sigma_{\rm soft}$ must stay positive or the "no resolved photon" branch has negative probability, and it goes negative beyond $v_{\rm max}/v_{\rm min} \sim 10^5$ at CLAS12 kinematics; `choose_vmin` takes the largest ratio that keeps the soft fraction above a floor and refuses to generate if there is none.

In one line: `subtract=True` computes the correction, `subtract=False` generates the events. One cross section, two ways of cutting it.

$\sigma_{\rm rad}$ is *not* proportional to $\sigma_{\rm Born}$ — and what that costs

The claim is *not* $\sigma_{\rm rad} \propto \sigma_{\rm Born}$. It is that $\sigma_{\rm rad}$ is a *linear, degree-1 homogeneous functional* of $\sigma_{\rm Born}$. Homogeneity means a rescaling of the model leaves η untouched; linear-but-non-local means η still depends on the model's *shape*. Both halves are needed, and they are different statements.

That it is not proportionality is settled by holding $\sigma_{\rm Born}$ at the point fixed — each perturbation equals unity there — and changing only the shape away from it:

perturbation	$\sigma_{\rm Born}$	σ_F	η
reference	8.3655e-05	−3.8327e-06	0.979770
SF $\times\, e^{-0.6(W^2-W_0^2)}$	8.3655e-05	+4.3160e-06	1.077180
SF $\times\, e^{+0.6(W^2-W_0^2)}$	8.3655e-05	−7.5031e-06	0.935894
SF $\times\, e^{-1.0(t-t_0)}$	8.3655e-05	−6.3523e-06	0.949651
SF $\times\, e^{+1.0(t-t_0)}$	8.3655e-05	−7.3283e-10	1.025577
SF $\times\, (Q^2/Q_0^2)^{+1}$	8.3655e-05	−1.3809e-06	1.009079

$\sigma_{\rm Born}$ frozen to all digits, η moving. At the steepest W slope σ_F even changes sign and η exceeds unity, because radiation then feeds events *into* the point from higher W^2. Were $\sigma_{\rm rad}$ proportional to $\sigma_{\rm Born}$, every row would read 0.979770.

The size, honestly — with the production model's own freedoms. Those perturbations are ad hoc: equal-percentage changes to each logarithmic slope are not what `jpsi.vmd2026` is free to do. The model is

$$\sigma_T = \sigma_\gamma(W)\left(1+\frac{Q^2}{m_{J/\psi}^2}\right)^{-k} \times \mathrm{dipole}(t; m_g^2), \qquad \sigma_\gamma(W) = A\,x^\alpha e^{-Bx}, \quad x = W - W_{\rm thr}, \tag{16}$$

with $k=2$ (VMD), $\Lambda^2 = m_{J/\psi}^2$, $m_g^2 = 1.78$ GeV^2, $c_R = 0.4$, $\sigma_\gamma(W)$ taken from `jpsi.roberts2026`, and validity $W^2 \in (16.4, 400)$, $Q^2 \in (0, 30)$, $|t| \in (0, 12)$.

The Q^2 shape is not a model freedom. σ_T is essentially flat in Q^2 across the FT box; the entire $1/Q^2$ of $d\sigma_{ep}$ is the virtual-photon flux:

Q^2	$(1+Q^2/9.590)^{-2}$	$\partial\ln(\mathrm{VMD})/\partial\ln Q^2$
0.001	0.99979	−0.00021
0.030	0.99377	−0.00624
0.100	0.97947	−0.02064
0.250	0.94984	−0.05081

Of the measured $\partial\ln\sigma_{\rm Born}/\partial\ln Q^2 = -1.008$ at $Q^2 = 0.03$, the flux gives −1.002 (exact QED, no freedom) and the model only −0.006. The "Q^2 slope" column of the ad hoc table is therefore meaningless for this model and is dropped.

Varying the model's *actual* parameters — m_g^2 by its published $\pm 4.8\,\%$ ($m_g^2 = 1.78$ from $m_s = 1.336 \pm 0.032$ GeV, GlueX PRC 108 025201 and CLAS12 2026), k over the whole range 1–3, and the two W-shape exponents by an assumed $\pm 30\,\%$:

Q^2	W^2	t	η_0	m_g^2 ±4.8%	$k : 1 \ldots 3$	α ±30%	B ±30%
0.03	17.50	−1.50	0.92671	0.00036	0.00002	0.00910	0.00131
0.03	19.50	−1.50	0.97977	0.00063	0.00009	0.00981	0.00334
0.03	20.50	−0.80	0.99025	0.00039	0.00025	0.00465	0.00254
0.25	19.00	−1.50	0.96881	0.00074	0.00050	0.01171	0.00345

So for the production model the hierarchy is

α (near-threshold rise of $\sigma_\gamma(W)$)	0.5–1.2 %	**dominant**
B (W fall-off)	0.1–0.3 %	
m_g^2 (t dipole, published error)	0.04–0.07 %	
k (VMD power, 1–3)	negligible	

— the W shape of the photoproduction cross section, not t and certainly not Q^2. An earlier revision concluded $t > Q^2 > W$; that followed from the ad hoc equal-percentage slopes and is wrong for `jpsi.vmd2026`. The $\pm 30\,\%$ on α and B is an assumption rather than a published error, while the m_g^2 number is published, so the 0.5–1.2 % is an upper bound of the right order rather than a measured uncertainty.

model normalisation wrong by any factor	effect on C^{rad}: **exactly zero**
model parameters within their freedom	effect on C^{rad}: $\sim$**1 %**, W-shape led

That $\sim 1\,\%$ is the number to carry in the systematics table against "model shape inside the bin", and step 7 reduces it further. The method is robust — but not because σ_{rad} is proportional to σ_{Born}. It is robust because η is scale-free and only mildly shape-sensitive.

Why the two forms are the same, exactly

Substituting ε and C^{rad} into Form B, n_i^{bdec} *cancels*:

$$\varepsilon_i^{\mathrm{Born}}\, C_i^{\mathrm{rad}} = \frac{n_i^{\mathrm{bdec}}}{m_i^{\mathrm{bdec}}} \cdot \frac{w^{\mathrm{rad}} n_i^{\mathrm{rad}}}{w^{\mathrm{bdec}} n_i^{\mathrm{bdec}}} = \frac{w^{\mathrm{rad}} n_i^{\mathrm{rad}}}{w^{\mathrm{bdec}} m_i^{\mathrm{bdec}}}, \qquad \bar{\sigma}_i^{\mathrm{Born}} = \frac{N_i^{\mathrm{data}}\, w^{\mathrm{bdec}} m_i^{\mathrm{bdec}}}{\mathcal{L}\,\mathcal{B}\,\Delta\Omega_i\, w^{\mathrm{rad}} n_i^{\mathrm{rad}}}. \tag{17}$$

Comparing with Form A gives the exact condition for the identity,

$$\boxed{\bar{\sigma}_i^{\mathrm{Born,model}} = \frac{w^{\mathrm{bdec}}\, m_i^{\mathrm{bdec}}}{\Delta\Omega_i}} \tag{18}$$

that is, the model's bin-averaged Born cross section taken from the *truth* counts of the Born Monte Carlo — generated events in the bin, times the cross section per event, per unit bin volume — rather than from an integral of the model. Use Eq. (18) and the two forms agree numerically, not merely algebraically. That is what "consistently" means.

Form A is what should be computed, for three reasons.

1. Acceptance, efficiency, resolution smearing, bin migration and radiation all sit inside one measured number, n_i^{rad}, obtained by running the same analysis code over reconstructed Monte Carlo. Nothing is factorised that does not factorise.

2. The $M(ee)$ window efficiency cancels. The same window is applied to the data and to $n_i^{\rm rad}$, so it never has to be computed, and the radiative lineshape is simulated rather than assumed.

3. **A wrong model normalisation cancels exactly.** If the model is scaled by a factor f, then $n_i^{\rm rad} \to f\, n_i^{\rm rad}$, hence $R_i \to R_i/f$ and $\bar\sigma_i^{\rm Born,model} \to f\,\bar\sigma_i^{\rm Born,model}$, so

$$\sigma_i^{\rm Born} = R_i\,\bar\sigma_i^{\rm Born,model} \quad \text{is invariant under} \quad \sigma^{\rm model} \to f\,\sigma^{\rm model}. \tag{19}$$

The third point is not academic, and the instructive case is the model this campaign *superseded*. `jpsi.diffrad_vm` carried $\sigma_T \propto Q^{-3.11}$, taking M. Tenorio's fit exponent of $d\sigma_{ep}/dQ^2dW$ — which already contains the flux $\sim 1/Q^2$ — and putting it into σ_T, so that σ_{γ^*} fell as $Q^{-3.1}$ instead of being flat and the model sat a factor 4–6 below the data at Forward Tagger kinematics. `jpsi.vmd2026` (VPK, 29.10.2026) fixed precisely that: pure VMD in Q^2, normalisation from the $\sigma_\gamma(W)$ fit of `jpsi.roberts2026`. For production the invariance is therefore insurance rather than a crutch — but it is the reason that episode would not have biased an extraction done this way. Form A is immune to that. What survives is only the *shape inside the bin*, and step 7 iterates it away.

3 Procedure

Step 1 — one cut list. Write the cuts once and apply them to the data, the radiative MC and the Born MC from the same code. Include the exclusivity cut. Both cards carry `vcut = 0`, so nothing was cut at generation and the whole radiative phase space is in the files; the exclusivity cut is applied *offline*, and that is the only way migration through resolution is simulated.

Step 2 — signal yield per bin. Count in a window around $M(ee) = 3.0969$ GeV. Do *not* fit a free lineshape: a free Gaussian width absorbs the radiative tail and biases the yield. Window counting reduces that model bias by about a factor 5. Subtract background from sidebands or a fitted continuum, and propagate the subtraction error into $N_i^{\rm data}$.

Step 3 — the same window on the MC. The identical $M(ee)$ window applied to the reconstructed radiative MC gives $n_i^{\rm rad}$. Both arms used here run `idecay = 2`, so the decay final-state radiation is simulated and the lineshape is never assumed. That is what makes the window efficiency cancel between the data and $n_i^{\rm rad}$, and what keeps the decay radiation *out* of $C^{\rm rad}$, where it would be double-counted against $\mathcal{B}$ (Sec. 5.2).

Step 4 — normalise. $Y_i^{\rm rad} = \mathcal{L}\,\mathcal{B}\,w^{\rm rad}\,n_i^{\rm rad}$. Average `sigma_nb` over jobs; all production jobs share the card, so $w^{\rm rad}$ is common to them.

Step 5 — form R_i and multiply by the model. Bin-average the model's Born cross section over the bin using the model's own density, *or* quote at the bin centroid with a bin-centring correction. State which. The two differ wherever the t-slope is steep, which near threshold is everywhere.

Step 6 — compute $C_i^{\rm rad}$ in any case. Form A does not need it, but it must be reported, and its bin-to-bin behaviour is the diagnostic that says whether the correction is sane. A $C^{\rm rad}$ that is flat in W, or above unity, means something upstream is wrong. The published 0.80/0.76/0.62 by W bin was measured with the `born` arm in the denominator, so it carries the decay-lepton factor as well; the extraction's $C^{\rm rad} = {\rm rad/borndecay}$ will sit closer to unity.

The third arm is *not* a free check on $F(K)$, as an earlier revision of this note claimed. Measured over 300 jobs per arm, borndecay/born $= 0.98879 \pm 0.01296$. Since $\int MB$ is identical for the two arms and

`written` = 5000 in both, that ratio is purely the effect of decay FSR on the *generation* daughter window ($|p| \in [1, 9]$ GeV, $\theta \in [5, 40]^\circ$). The analysis cut $|p| > 1.95$ GeV is tighter, so it is $F(K)$ for a window the analysis does not use, and at $1.1 \pm 1.3\,\%$ it is consistent with nothing happening. The decay-radiation model is tested against the PDG instead, Sec. 5.3.

Step 7 — iterate. Reweight the Monte Carlo event by event by $\sigma^{\text{extracted}}/\sigma^{\text{model}}$ as a function of (W, Q^2, t), then recompute n_i^{rad} and R_i. Two or three passes converge. This is required rather than cosmetic: the bin-averaging inside wide bins and the migration across bin edges both carry the model's shape, and the dominant freedom is the near-threshold W rise of $\sigma_\gamma(W)$ (Sec. 2) — exactly where the J/ψ data are. It is *not* about a Q^2 fit range: `jpsi.vmd2026` is valid over $Q^2 \in (0, 30)$, far beyond the box.

Step 8 — efficiency scale factors. GEMC does not reproduce the Forward Tagger electron or the tracking efficiencies. Apply the measured data/MC ratios to Y_i^{rad}, per bin wherever they are kinematics-dependent.

Step 9 — quote it. For $ep \to e'p'J/\psi$ no flux factor is needed. If converting to $\sigma(\gamma^* p \to J/\psi\, p)$, state the flux convention (Hand or Gilman) explicitly: it moves the number.

4 Statistical error

From Form A, with the three counts independent,

$$\left(\frac{\delta R_i}{R_i}\right)^2 = \frac{1}{N_i^{\text{data}}} + \frac{1}{n_i^{\text{rad}}} + (\text{background subtraction term})\,. \tag{20}$$

The `borndecay` arm enters the statistical error *only* through C^{rad} and $\varepsilon^{\text{Born}}$, that is only in Form B and in the reported diagnostics. That is not a reason to generate less of it.

5 Facts checked in the actual samples, 6 November 2026

5.1 The branching ratio is not in the generator

```
grep -rniE "branching|0.0597|5.97" --include=*.py exclurad_py/   ->   nothing
```

`report.sigma_nb` is the **full J/ψ production cross section**. Every event is decayed to e^+e^- with probability one (`idecay = 2`). The data yield is an e^+e^- yield. Therefore $\mathcal{B}$ multiplies the Monte Carlo, once. Omitting it makes the answer wrong by 16.7.

5.2 Which radiative processes are corrected, and which are not

The Born card of `prod_jpsi_f18in_20261009` states the rule in its own words (2026-10-21):

> THE BORN ARM MUST NOT RADIATE ... The rule is that whatever is present in BOTH arms cancels in $N_{\text{rad}}/N_{\text{born}}$ and is therefore NOT corrected for; only what differs between them is removed. The published number is σ_{Born} at a definite beam energy, so the external radiator belongs in the radiative arm and must be absent here. Left on in both arms — as it was until 2026-10-21 — the ratio returns a Born cross section smeared over the beam-energy distribution in the target, which nothing can be compared against.

Read as a question — *is this process already counted in something we normalise by?* — it settles all three fields at once:

process	field	already counted?	corrected?	rad / Born
vertex radiation	`iborn`	no	yes	0 / 1
pre-vertex straggling	`itarg`	no	yes	1 / 0
decay-lepton radiation	`idecay`	**yes, in $\mathcal{B}$**	**no**	2 / 2

Decay radiation is inside the PDG branching ratio (Sec. 5.3), so it must *cancel* and therefore sit in both arms. Straggling is inside nothing, and the published σ is at a definite beam energy of 10.6 GeV, so it must be *removed* and therefore sit in the radiative arm only. The two questions have opposite answers and are not in conflict.

Sizes. The `itarg` term was measured with a dedicated `itarg` = 1 Born run: σ falls by 1.24 % (0.345396 against 0.349740); see `docs/rc_method_jpsi/`. The straggling itself comes in two populations that must not be confused:

per TRIAL (card, at generation)	mean loss 0.34 %; 2.54 % of trials lose > 0.1 %, 1.70 % > 1 %
per WRITTEN event (2000-event pdet sample)	mean loss 0.0264 %, max 6.1 %

The acceptance suppresses the mean by about 14×, because $y \in [0.76, 0.99]$ and the threshold at $W^2 = 16.283$ kill the events that lost beam energy. Quote the right one.

And `iborn` = 1 with `itarg` = 0 and `idecay` = 2 is exactly the `borndecay` arm, the subject of the next subsection.

5.3 The Born arm must be `borndecay`, because $\mathcal{B}$ is the full width

This is the crux of the whole procedure (VPK, 6 November 2026).

The normalisation uses $\mathcal{B}(J/\psi \to e^+e^-) = (5.971 \pm 0.032)$ % from the PDG. That number is the *radiation-inclusive* leptonic width: $\Gamma(J/\psi \to e^+e^-)$ is measured inclusively, so $J/\psi \to e^+e^-\gamma$ is already counted inside it.

Radiation of the decay leptons is therefore *not* a correction to be removed. It is part of the physical final state that $\mathcal{B}$ counts. Its only effect on the measurement is on the *acceptance* — through the $M(ee)$ window and through the lepton momenta — so it must be simulated in every arm that is used, so that it cancels in $C^{\rm rad}$ and is folded correctly into ε. Put it into $C^{\rm rad}$ instead and it is counted twice: once in $\mathcal{B}$, once in the correction.

The Born arm for an extraction is therefore the one with Born *production* and decay radiation *on*. The `osg_mariana_lund` campaign carries exactly that, as a third arm. Card fields verified on ifarm, 6 November 2026:

field	switches	`rad`	`born`	`borndecay`
1	model	`jpsi.vmd2026_pdet`		
4	`iborn` — vertex radiation	0	1	1
8	`vcut`	0.0	0.0	0.0
12	`idecay` — decay-lepton radiation	2	1	2
18	`itarg` — pre-vertex radiator	1	0	0

$$C^{\rm rad} = \frac{\text{rad}}{\text{borndecay}}: \quad \text{production radiation only (vertex + pre-vertex straggling), decay radiation cancels.} \tag{21}$$

Statistics: 2000 LUND files of 5000 events per arm per period (fa18 / sp18 / sp19), and all three arms are linked into each of the nine OSG dataset directories, so `born`, `borndecay` and `rad` are all submitted.

The `born` arm (`idecay` = 1) is *not* wrong. It was built for the RC-method comparison of `docs/rc_method_jpsi/`, where method 2 multiplies the analytic decay-lepton factor $F(K)$ explicitly, so keeping the decay radiation inside $C^{\rm rad}$ was deliberate and consistent there. It is the wrong denominator for a cross-section extraction, which is a different question.

The model differs between the two campaigns: `osg_mariana_lund` uses `jpsi.vmd2026_pdet` while `prod_jpsi_f18in_20261009` uses `jpsi.diffrad_vm_pdet`. Form A is invariant under the model normalisation, so that costs nothing there; only the shape inside the bin matters, and step 7 iterates it away.

5.4 The decay-radiation model agrees with the PDG radiative branching fraction

The premise of Sec. 5.2 — that $\mathcal{B}$ is radiation-inclusive — is in the PDG listing itself, and the listing also supplies a number the generator can be tested against. PDG 2025, $J/\psi(1S)$, with $e^+e^-\gamma$ *indented* under e^+e^-, which in PDG notation marks a submode, i.e. a subset of the line above:

	mode		branching fraction	
Γ_5	e^+e^-		(5.971 ± 0.032)	%
Γ_6	$\quad e^+e^-\gamma$	[a]	(8.8 ± 1.4)	$\times 10^{-3}$
Γ_7	$\mu^+\mu^-$		(5.961 ± 0.033)	%

[a] For $E_\gamma > 100$ MeV.

So $\mathcal{B}(e^+e^-)$ is the full width and the radiative part is quoted as a fraction of it:

$$\frac{\Gamma_6}{\Gamma_5} = \frac{8.8 \times 10^{-3}}{5.971 \times 10^{-2}} = 14.74 \pm 2.35\,\% \qquad \text{for } E_\gamma > 100 \text{ MeV}. \tag{22}$$

Since $M_{ee}^2 = M^2 - 2Mk^*$, that threshold is $M(ee) < 2.99523$ GeV. Calling our own decay model directly, with no production and no acceptance cut:

`DecayFSR` (`idecay` = 2, exact ME), 300 000 decays	fraction
$E_\gamma > 20$ MeV	23.95 %
$E_\gamma > 50$ MeV	18.63 %
$\mathbf{E_\gamma > 100}$ **MeV**	$\mathbf{14.38 \pm 0.06}$ %
$E_\gamma > 200$ MeV	10.12 %
$E_\gamma > 400$ MeV	6.06 %
$E_\gamma > 800$ MeV	2.56 %
PDG, same threshold	$\mathbf{14.74 \pm 2.35}$ %
difference	$-0.36\,\% = 0.15\sigma$

Mean $E_\gamma = 72.0$ MeV, maximum 1548 MeV. This is the first independent check on `generator/decay_rad.py`. It also explains the card's own remark that "$\sim$15 % of decays end below $M(ee) = 3.0$ GeV": 3.0 GeV corresponds to $k^* = 95.4$ MeV, essentially the PDG threshold, which is why the two numbers coincide.

5.5 The card header of `prod_jpsi_f18in_20261009` is false

That campaign's radiative card states "Born twin: IDENTICAL except iborn (field 4)" and offers a `diff` command to prove it. The command prints five differences: fields 4, 11 (seed), 12 (`idecay`), 13 (`outroot`) and 18 (`itarg`). The danger is not the confusion but the repair: a reader who trusts the comment may "fix" field 12 or field 18 to make the diff clean, and a Born arm whose `itarg` or `idecay` has been flipped silently changes what $C^{\rm rad}$ means. It is the comment that is wrong, not the cards.

That campaign's Born arm also has `idecay = 1`, which is why it is not the arm to use for an extraction (Sec. 5.2). Its `itarg = 0` is nevertheless safe from the point-vertex bug: the fix `bbe6628` (2026-10-22 11:25) is an ancestor of the sample's recorded commit `8ac2fc0` (2026-10-22 12:36), so v_z is spread over the 5 cm cell rather than pinned to zero.

5.6 C^{rad} cannot be taken from `sigma_nb`

arm	`report.sigma_nb`
born	3.494470×10^{-4}
rad	3.368744×10^{-4}
ratio rad/born	0.9640

`sigma_nb` counts events *before* the acceptance preset, so any rad/Born ratio built from it is about 1.33 % high, and in any case it is a box average rather than a bin correction. C_i^{rad} must come from reconstructed yields inside the bin.

5.7 The `prod_jpsi_f18in_20261009` arms are not at equal statistics

	born	rad
`out_inb`	1120	1167
`out_outb`	852	1180

The Born outbending arm is 28 % short, which caps the precision of both C^{rad} and $\varepsilon^{\mathrm{Born}}$. It should be finished before a systematic is quoted on either.

5.8 `iacc = jpsi` is a hard generation cut, not a flag

`events.py` executes `continue` on failure: the event is never written. The windows travel with the sample in the `.lund.info` sidecar:

particle	$\lvert p\rvert$ (GeV)	θ (deg)
proton	0.2–4.5	3–45
daughter	1.0–9.0	5–40
electron	0.1–2.5	2–20

If any analysis cut is *looser* than the corresponding preset edge, the Monte Carlo has exactly zero events there and the acceptance is wrong, not merely imprecise. Measured in a 2000-event sample, the live edges are the proton θ floor (minimum 3.013°, zero events below) and the daughter θ ceiling (maximum 39.96°). The proton $|p|$ window never bites: the kinematic minimum is 0.7156 GeV/c against a 0.2 GeV/c floor.

5.9 One `sigma_nb` convention across two build commits — a false alarm, closed

In `osg_mariana_lund` the arms were not all built from one commit: `born` and `rad` at `8377340` (2026-10-29), `borndecay` at `3648a81` (2026-10-30). The *only* change to the generator between those commits is the `sigma_nb` normalisation fix `c10d09e`, "`sigma_nb` normalises with the events that reached the LUND file". Before it the numerator was the stage-1 accept counter, which includes events the `iacc` daughter preset then drops — on the J/ψ preset about 1.70× too large. Had `rad` carried the bug and `borndecay` not, C^{rad} = rad/borndecay would be wrong by that factor. **It is not.**

The test: σ_{nb} = (numerator/trials) × $\int MB$, and $\int MB$ is a deterministic table integral over the box, so it must come out *constant* across the jobs of one arm. Backing it out both ways, 60 jobs per arm:

arm	commit	jobs	mean $\int MB$	rel. spread	numerator	
`rad`	`8377340`	60	3.821439e-03	3.4e-16	`written`	constant
			2.224601e-03	4.3e-02	pre-preset	varies
`born`	`8377340`	60	2.855625e-03	3.0e-16	`written`	constant
`borndecay`	`3648a81`	60	2.855625e-03	3.0e-16	`written`	constant

All three use the post-preset numerator, and the decisive line is the last two: `born` and `borndecay` back out the *identical* $\int MB = 2.855625 \times 10^{-3}$ to machine precision — as they must, since the Born majorant cannot depend on `idecay`. The `8377340` sidecars were corrected by the conductor's `campaign_lib` at write time (`a6240da`), `borndecay` carries the correction natively (`c10d09e`), and `bce9e23` stops `campaign_lib` double-correcting a sidecar that already has it.

Nothing to correct, nothing to regenerate. Recorded because the alarm was the author's own and the check is cheap: if a campaign ever mixes commits again, back out $\int MB$ per arm and demand that it be constant.

5.10 t from the proton, and the choice of t or t'

The proton t estimator has RMS 0.047 GeV^2 against 0.230 for the J/ψ one, a factor 5–6 better. The existing analysis ntuple carries `t_prime`, not `t`. Since $|t|_{\min}$ runs from 2.10 GeV^2 at threshold to 0.4536 at the top of the W range, bins in t' and in t map onto appreciably different regions. Fix the choice once and use it in the data, in both Monte Carlo arms, and in the model bin-averaging.

For reference, the recoil proton follows from t alone through $t = (p - p')^2 = 2M^2 - 2ME_p$, so

$$E_p = M + \frac{|t|}{2M}, \qquad |\mathbf{p}_p| = \sqrt{|t|}\,\sqrt{1 + \frac{|t|}{4M^2}}. \tag{23}$$

At the kinematic ceiling $W^2 = M^2 + 2ME_{\text{beam}} = 20.7717$ GeV^2 ($W = 4.5576$ GeV),

$$|t|_{\min} = 0.4536 \text{ GeV}^2, \qquad |\mathbf{p}_p| = 0.7156 \text{ GeV}/c, \qquad \theta_p = 0.67^\circ, \tag{24}$$

so at the forward edge the recoil proton travels down the beamline and is undetectable; the 3° preset floor removes it. Over the whole box the recoil proton never exceeds $\theta_p = 25.3^\circ$: it is a Forward Detector particle everywhere, never a Central Detector one.

6 Systematics

Source	How to estimate
model shape inside the bin	spread over iterations; vary $\sigma_\gamma(W)$'s exponents and m_g^2 within their freedom. Measured: $\sim$1 % on C^{rad}, dominated by the near-threshold W rise (Sec. 2)
exclusivity cut	vary the MX^2 window; C^{rad} is the quantity that moves
signal extraction	window width; sideband versus fitted background
MC statistics	n^{rad} and n^{bdec} separately
efficiency scale factors	their own uncertainties, bin dependent
luminosity	from the data side
branching ratio	0.54 % relative
bin centring	centroid versus bin-averaged model

Table 2: Systematic uncertainties to carry through Eq. (3).

7 Ntuple branches the chain needs

The reader in `production/analysis/campaign_plots.py` uses

```
E_photon, Q2, W, t_prime, invariantMass,
scattered_theta, scattered_p, proton_theta, proton_p,
positron_theta, positron_p, electron_theta, electron_p,
time_ee, x_FT, y_FT
```

The data and every Monte Carlo arm must expose the same set, filled by the same code, or step 1 cannot be honoured.

Companion notes

- `docs/RC_METHOD_FOR_JPSI.md` — how the π^0 RC-method study was done and what to change for J/ψ.
- `docs/RC_METHOD_COMPARISON.md` — MC ratio versus the analytic `rc_factor`, and what the difference costs.
- `docs/rc_method_jpsi/rc_method_jpsi_report.tex` — the measured C^{rad} for this campaign, both methods.
- `docs/BORN_XSEC_EXTRACTION_JPSI.md` — the Markdown form of this note.